

Proof-Grounded Patient-Specific Clinical Explanations from Knowledge-Graph Reasoning

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Abstract—Clinical decision-support systems may provide predictions or scores without preserving an auditable connection among patient observations, encoded medical knowledge, derived conclusions, and generated recommendations. We present the CKG Clinical Explanation Engine, a patient-specific proof-grounded explanation layer operating downstream of a training-free clinical knowledge-graph reasoner. The engine transforms frozen patient-level inference outputs and disease-specific knowledge representations into typed logical facts, rule-application traces with explicit premises, provenance-linked conclusions, and policy-licensed recommendations. A central semantic distinction separates measurement availability, representational completeness, and disease-specific activation: an observed measurement with zero diagnostic activation is not missing, while partially represented evidence remains distinct from genuinely unobserved evidence. Optional language generation is restricted to clinical content already licensed by the symbolic engine.

The framework was evaluated without modifying the underlying diagnostic reasoner. A full-corpus audit covered 6,720 patient records, 20,160 patient–disease traces, and 1,021,440 feature-evidence rows across five clinical reasoning workbooks. Information-Gate range validity and knowledge-source provenance were 100% (1,021,440/1,021,440); numerical cross-sheet fidelity was 100% (120,960/120,960); and exported logical and deterministic-report trace completeness were both 100% (20,160/20,160). Evidence-state consistency was 99.7028% (1,018,404/1,021,440; 95% CI 99.6920–99.7131), with 3,036 inconsistencies localized to two source workbooks. The corpus additionally contained 86,783 observed feature instances with zero activation and 139,949 observed instances with partial representation. A separate end-to-end audit of 25 patient reports completed without execution failures and passed all pre-specified proof-completeness, recommendation-licensing, knowledge-source, and evidence-state checks. These structural and implementation audits demonstrate a machine-auditable route from knowledge-graph inference to patient-specific clinical explanation while preserving explicit boundaries among diagnostic inference, symbolic justification, and language realization.

Index Terms—clinical decision support, explainable artificial intelligence, knowledge graphs, symbolic reasoning, first-order logic, natural deduction, neuro-symbolic AI, clinical explanation

I. INTRODUCTION

Clinical artificial intelligence increasingly combines statistical models, structured medical knowledge, and natural-language interfaces. However, producing a clinically plausible statement is fundamentally different from establishing that the statement follows from available patient observations and identifiable medical knowledge. This distinction becomes particularly important when generated reports contain diagnostic interpretations, measurements, tests, or clinical recommendations.

Large language models (LLMs) have demonstrated substantial medical knowledge and question-answering ability [6]. Nevertheless, performance on medical questions does not establish reliable autonomous clinical reasoning. Evaluations in realistic clinical settings have reported difficulties with information gathering, guideline adherence, laboratory interpretation, and workflow integration [7]. Recent studies have also identified limitations in the ability of LLMs to recognize insufficient knowledge or missing information [8]. Consequently, linguistic fluency alone cannot establish evidential provenance or logical validity.

Symbolic systems provide a complementary capability because premises, knowledge representations, and inference rules can remain explicit. Recent neuro-symbolic clinical research has demonstrated the value of coupling language models with rule-based verification to obtain deterministic and auditable outputs [9]. The question addressed here differs slightly: rather than allowing an LLM to perform the primary clinical reasoning, can the clinically relevant content first be licensed by a patient-specific symbolic proof and only subsequently verbalized?

Our upstream CKG Reasoner follows a training-free, ontology-grounded paradigm in which patient observations are mapped to disease-specific Evidence Feature Nodes (EFNs), followed by patient–reference matching, symbolic–probabilistic evidence accumulation, disease-level correspondence, ranking, uncertainty handling, and deterministic rules [1]. The present work addresses the downstream explanation problem. Given an existing patient-specific inference state, the objective is to generate an auditable clinical explanation without retraining, recalibrating, or modifying the diagnostic mapping.

The contributions of this study are fourfold:

- 1) a patient-specific logical state that explicitly separates observed, partially represented, and genuinely unobserved evidence from disease-specific activation;
- 2) transformation of CKG patient outputs and disease YAML knowledge into typed first-order predicates and inspectable rule-application traces;
- 3) a closed recommendation policy in which every clinical action requires a symbolic antecedent and identifiable provenance; and
- 4) a reproducible frozen-system evaluation over 6,720 patients, 20,160 patient–disease traces, and more than one million feature-evidence records.

An optional LLM is placed only after the symbolic licensing stage and is not permitted to independently introduce diagnoses, measurements, tests, medications, or recommendations.

II. RELATED WORK

Explainable artificial intelligence encompasses post-hoc explanation, intrinsically interpretable modeling, causal explanation, and explicit rule-based reasoning [4], [5]. These paradigms address different questions. In clinical decision support, an additional requirement is whether a statement can be traced to a patient observation, knowledge source, and identifiable inference procedure.

Explicit medical reasoning predates modern machine learning. MYCIN demonstrated that structured medical knowledge and uncertainty could be represented computationally [2]. Natural deduction provides a related formal foundation in which a conclusion follows from explicit premises through identifiable inference rules [3].

Modern LLMs create a different opportunity and risk. Medical LLMs can achieve high performance on knowledge-oriented tasks [6], yet realistic clinical evaluations demonstrate that correct final answers do not necessarily imply reliable clinical reasoning [7]. Griot *et al.* further showed substantial limitations in metacognitive behavior, including recognizing when sufficient information is unavailable [8].

Hybrid approaches therefore increasingly combine neural flexibility with symbolic constraints. Prenosil *et al.* coupled GPT-4-based information extraction with a rule-based expert system and demonstrated auditable reasoning over clinical imaging reports [9]. Our architecture shares the objective of audibility but places the boundary differently: the symbolic CKG system already determines the patient-specific clinical state, and optional language generation is restricted to verbalizing pre-authorized claims. Thus, the LLM is not the source of diagnostic reasoning.

III. CKG CLINICAL EXPLANATION ENGINE

A. Relationship to the Upstream Reasoner

The explanation engine operates downstream of CKG Reasoner. For patient p and candidate disease D , upstream outputs may contain positive and negative evidence, normalized evidence, disease-level similarity, disease score, feature-level Information Gate (IG) values, and Diagnostic Evidence Coverage (DEC).

The explanation layer treats these quantities as facts describing the upstream inference state. It does not fit, optimize, recalibrate, or alter the diagnostic score. The processing chain is

$$\begin{aligned}
 &\text{patient data} \rightarrow \text{CKG inference} \\
 &\quad \rightarrow \text{typed facts} \\
 &\quad \rightarrow \text{proof state} \\
 &\quad \rightarrow \text{licensed claims/actions} \\
 &\quad \rightarrow \text{clinical report.}
 \end{aligned} \tag{1}$$

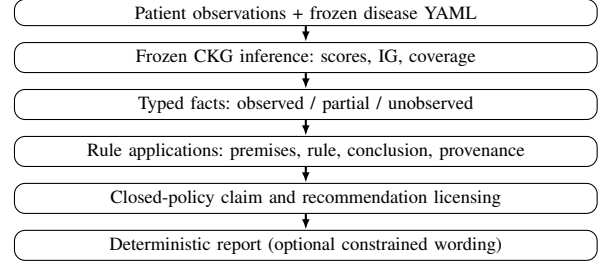


Fig. 1. Separation of frozen diagnostic inference, symbolic authorization, and downstream report realization. Optional language generation was not experimentally evaluated.

DEC is interpreted as evidence completeness rather than disease probability.

B. Observed, Partial, and Unobserved Evidence

A central design requirement is separation of measurement availability from disease-specific activation.

Let $m_{p,f}$ indicate whether the measurement required for feature f is available for patient p , and let $a_{p,f} \in [0, 1]$ denote the fraction of the feature's expected components that can be evaluated. We define

$$\text{Observed}(p, f) \iff m_{p,f} = 1, \tag{2}$$

$$\text{Partial}(p, f) \iff m_{p,f} = 1 \wedge 0 < a_{p,f} < 1, \tag{3}$$

$$\text{Unobserved}(p, f) \iff m_{p,f} = 0. \tag{4}$$

Disease-specific patient-reference activation is represented separately:

$$\text{IG}(p, f) \in [0, 1]. \tag{5}$$

Therefore,

$$\text{IG}(p, f) = 0 \not\Rightarrow \text{Unobserved}(p, f), \tag{6}$$

and

$$\text{Partial}(p, f) \not\Rightarrow \text{Unobserved}(p, f). \tag{7}$$

These distinctions are important because an available laboratory measurement can be weakly or non-supportive for a particular disease while remaining an observed measurement. Likewise, partial representation concerns the mapping between an available measurement and the components expected by the knowledge model; it does not imply that the patient measurement itself is absent.

C. Knowledge-Grounded Predicates

Patient and knowledge states are represented through predicates such as

$$\text{PatientValue}(p, x, v), \quad (8)$$

$$\text{MappedFeature}(x, f), \quad (9)$$

$$\text{Role}(f, D, \rho), \quad (10)$$

$$\text{IG}(p, f, g), \quad (11)$$

$$\text{Cov}(p, f, a). \quad (12)$$

Here ρ denotes the encoded diagnostic role, such as Pathogenic, Hallmark, Major, or Supportive. Roles and disease-specific relationships are obtained from the packaged knowledge representation rather than assigned by a language model.

IV. PROOF-GROUNDED REASONING

Let Γ_p denote the available facts for patient p . The rules below express the intended inference semantics. The evaluated completeness checks verify the presence of exported premises, rule identifiers, conclusions, and reasons; they do not constitute independent formal verification of logical soundness or clinical truth.

An observed feature with an encoded diagnostic role and meaningful activation may license an evidence-support statement:

$$\frac{\text{Observed}(p, f) \quad \text{Role}(f, D, \rho) \quad \text{IG}(p, f, g) \quad g \geq \tau}{\text{EncodedSupport}(p, D, f)} R_{\text{sup}}. \quad (13)$$

Observed evidence with incomplete component coverage instead derives

$$\frac{\text{Observed}(p, f) \quad \text{Cov}(p, f, a) \quad 0 < a < 1}{\text{PartiallyRepresented}(p, f, a)} R_{\text{part}}. \quad (14)$$

No missing-evidence conclusion follows from R_{part} .
For genuinely unavailable critical evidence,

$$\frac{\text{Unobserved}(p, f) \quad \text{Role}(f, D, \rho) \quad \rho \in \mathcal{R}_{\text{critical}}}{\text{CriticalEvidenceIncomplete}(p, D, f)} R_{\text{miss}}. \quad (15)$$

A clinical test is licensed only if the unresolved evidence is associated with an explicitly catalogued action:

$$\frac{\text{CriticalEvidenceIncomplete}(p, D, f) \quad \text{LicensedTest}(D, f, t)}{\text{Recommend}(p, t)} R_{\text{test}}. \quad (16)$$

Thus, low coverage, partial representation, or an LLM suggestion cannot independently create a clinical test recommendation.

A. Constrained Language Realization

The deterministic renderer is the evaluated report-generation mechanism. An optional, unevaluated LLM can be placed downstream of the rule engine, but receives licensed proof content rather than authority to generate new clinical facts.

The output verifier imposes the following constraints:

- 1) clinical statements must correspond to licensed claims;

- 2) numerical values must originate from patient or CKG facts;
- 3) recommended actions must exist in the controlled policy vocabulary; and
- 4) medication and dose generation are disabled.

This separates clinical-content authorization from linguistic realization.

V. EXPERIMENTAL EVALUATION

A. Frozen-System Protocol

Evaluation was performed after freezing the clinical explanation package. A separate experimental harness inspected existing CKG reasoning workbooks and invoked the patient-specific engine without changing diagnostic scores, knowledge files, thresholds, or reasoning rules.

Five usable workbooks contained 6,720 patient records. Across three candidate diseases per patient, the deterministic audit covered 20,160 patient–disease traces and 1,021,440 feature-evidence rows. An additional influenza workbook supplied to the evaluation pipeline was zero-byte and was therefore recorded as unusable and excluded before analysis.

Two evaluation levels were used:

- 1) a full-corpus audit of exported reasoning structures; and
- 2) an end-to-end audit using the frozen CKGPatientReasoner.

B. Full-Corpus Audit

The deterministic audit evaluated evidence-state consistency, IG range validity, representation coverage, component consistency, feature-to-YAML provenance, numerical agreement between workbook structures, exported logical trace completeness, and deterministic-report completeness. These are structural invariants, not independent proofs of medical validity.

The experiment additionally counted two semantic stress conditions: observed evidence with $\text{IG} = 0$, and observed evidence with partial representation. These directly test Eqs. (6) and (7) on real exported inference traces.

C. End-to-End Engine Audit

A reproducibly seeded sample of five patients from each usable workbook was processed using the frozen patient reasoner, giving 25 end-to-end cases.

For each case, the audit required:

- 1) every proof step to contain premises, rule, conclusion, and reason;
- 2) every recommendation identifier to belong to the closed policy;
- 3) every referenced YAML knowledge source to exist; and
- 4) partial and unobserved evidence states to remain disjoint.

Execution exceptions and invariant failures were retained rather than removed.

TABLE I
FULL-CORPUS EXPLANATION-SYSTEM VALIDATION.

Criterion	Valid/evaluated	Rate
Evidence-state consistency	1,018,404/1,021,440	99.7028%
IG range validity	1,021,440/1,021,440	100%
YAML feature provenance	1,021,440/1,021,440	100%
Numerical cross-sheet fidelity	120,960/120,960	100%
FOL trace completeness	20,160/20,160	100%
Report trace completeness	20,160/20,160	100%

D. Statistical Reporting

For each invariant,

$$R_{\text{valid}} = \frac{N_{\text{valid}}}{N_{\text{evaluated}}}. \quad (17)$$

Binomial proportions are accompanied by 95% Wilson confidence intervals where informative. Feature-level, trace-level, numerical, and patient-level tests retain their respective denominators rather than being combined into a single accuracy measure.

VI. RESULTS

A. Corpus-Level Validation

Table I summarizes the principal audit results.

Evidence-state consistency was 99.7028% (95% CI 99.6920–99.7131). The audit identified 3,036 inconsistent exported feature-state records. These were localized rather than corpus-wide: 2,036 occurred in the 1,018-record dengue workbook and 1,000 in the 1,000-record dengue workbook; the other three usable workbooks contained no corresponding evidence-state inconsistency. The available audit identifies the affected exports but does not establish the underlying cause of these discrepancies or demonstrate that all affected patient explanations would remain unchanged. Root-cause analysis and re-export verification remain necessary.

The corpus contained 86,783 observed feature instances with $IG = 0$. These records provide direct empirical evidence that zero disease-specific activation is a frequent condition rather than an artificial edge case.

A further 139,949 observed feature instances had partial representational coverage, demonstrating that partial representation likewise requires an explicit state rather than being collapsed into missingness.

B. End-to-End Engine Audit

All 25 sampled patient reports completed without an execution exception. Every evaluated case passed the four pre-specified engine-level checks.

No end-to-end audit failure was recorded.

C. Patient-Specific Proof Example

Patient 152 illustrates the semantics of the system. For the Dengue Fever candidate, the upstream reasoner produced an evidence probability of 0.1179, Hallmark/Major cosine similarity of 0.7062, relative-magnitude similarity of 0.6737, disease score of 0.4993, and evidence coverage of 0.3715.

TABLE II
END-TO-END FROZEN-ENGINE AUDIT.

Criterion	Valid/evaluated	Rate
Successful execution	25/25	100%
Proof completeness	25/25	100%
Recommendation licensing	25/25	100%
YAML-source resolution	25/25	100%
Evidence-state disjointness	25/25	100%

The coverage value is explicitly interpreted as evidence completeness rather than disease probability.

Hemoconcentration and mild thrombocytopenia were observed Hallmark features with IG values of 1.000 and approximately 0.953, respectively. Leukopenia and neutropenia were also observed but had $IG = 0$. The latter therefore remained observed evidence rather than being reclassified as missing.

The same high-role features were partially represented because only part of their expected knowledge components was evaluable. In contrast, NS1 and PCR confirmation evidence were genuinely unobserved.

The natural-deduction trace first derived PartiallyRepresentedHighRoleEvidence from the observed measurements and their component coverage. A separate rule derived UnresolvedUnobservedEvidence from genuinely unobserved confirmation features. Only after this derivation did the policy license NS1 and PCR testing. The generated report therefore distinguishes incomplete representation from unavailable patient evidence and preserves provenance for the recommendation.

VII. DISCUSSION

The results demonstrate three important properties of the proposed architecture.

First, observation and activation cannot be treated as equivalent concepts. Across the evaluated corpus, 86,783 observed feature instances had zero Information-Gate activation. An implementation that interpreted zero activation as missingness would therefore change a substantial number of patient-specific explanations.

Second, partial representation is also a substantive implementation state. The 139,949 observed partial-representation instances demonstrate why availability of a patient measurement and completeness of its knowledge-model representation require separate semantics.

Third, provenance and numerical fidelity remained intact across the audited outputs. All evaluated feature records resolved to their packaged YAML knowledge representation, all 120,960 numerical cross-sheet comparisons were consistent, and all 20,160 exported logical and deterministic-report traces passed the corresponding structural completeness tests.

The 3,036 evidence-state inconsistencies are important rather than inconvenient results. They were retained in the denominator and localized to two exported source workbooks. In contrast, all 25 sampled executions of the current frozen patient reasoner preserved evidence-state disjointness; this

small sample does not rule out the same defect in other executions. We therefore distinguish corpus-export consistency from current end-to-end engine behavior rather than retrospectively removing the affected records to obtain a perfect result. This also illustrates a practical benefit of the audit architecture: representation inconsistencies can be exposed explicitly instead of being hidden inside a fluent clinical narrative.

The work is related to recent neuro-symbolic clinical systems that use rule-based reasoning to constrain or verify neural outputs [9]. However, our design places symbolic authorization before language generation. The LLM, when used, is not asked to discover the clinical conclusion. This boundary is motivated by evidence that strong medical-language performance does not guarantee reliable handling of uncertainty, missing information, or real clinical workflows [7], [8].

Several limitations remain. Logical validity does not prove that every encoded medical rule, threshold, or policy action is clinically correct; the system inherits limitations of its knowledge representation. The present evaluation is cross-sectional and does not validate temporal prognosis. Furthermore, software-level proof and provenance validation do not establish improved patient outcomes or clinician decision quality. Prospective and clinician-centered studies are therefore required before claims of clinical utility can be made.

The 25-case engine audit is also smaller than the corpus-level structural audit because complete patient-specific reasoning is computationally more expensive than validating exported traces. It should therefore be interpreted as an implementation-level complement to, rather than a replacement for, the full-corpus audit.

Finally, this study does not claim that the constrained architecture outperforms unrestricted LLM generation. A direct comparison among unconstrained generation, proof-context free-text generation, and identifier-constrained generation would require a fixed model and version, frozen prompts, repeated generations, and pre-specified human adjudication. That constitutes a separate validation question.

VIII. CONCLUSION

We presented a proof-grounded clinical explanation architecture operating downstream of training-free knowledge-graph reasoning. The framework explicitly separates measurement availability, representational completeness, and disease-specific activation; constructs inspectable rule-application traces; restricts clinical recommendations to policy-licensed actions; and places optional language generation after symbolic content authorization.

Evaluation across 6,720 patient records, 20,160 patient-disease traces, and 1,021,440 feature-evidence records demonstrated complete IG range validity and knowledge provenance, complete numerical cross-sheet fidelity, and complete exported logical/report trace completeness in their respective audits. Evidence-state consistency was 99.7028%, with the detected inconsistencies localized to two exported workbooks. A separate 25-patient frozen-engine audit completed without failures

across all pre-specified proof, provenance, recommendation, and evidence-state checks.

These results concern structural fidelity and implementation checks, not formal proof of clinical correctness or improved patient outcomes. They support treating clinical explanation as an auditable computational product rather than an unrestricted narrative. Equally importantly, the framework exposes inconsistencies when they occur, preserving a clear boundary between what was observed, what the knowledge representation can express, what the reasoning engine derives, and what a downstream language generator is permitted to say.

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